

A Horse Clinic chain called MTU Thoroughbreds Clinic wishes to hire you to design and implement a new database system for their expanding Horse care business. MTU Thoroughbreds runs a number of horse clinics around Ireland. A particular clinic will have many vets who is a member of staff. A member of staff manages at most one clinic (not all staff manage clinics). Each clinic in the MTU Thoroughbreds clinics chain will have a unique clinic number (clinicNo) along with address, phone number. Each member of staff will have a unique staff number (staffNo). Further staff details such as first name, last name, address, DOB, salary, position will also be stored. When a Horse owner contacts a clinic, the owner's Horse is registered with the clinic. An owner can own more than one Horse, but a Horse can only register with one clinic. Each owner has a unique owner number (ownerNo), name, address, phone number and each Horse has a unique Horse identifier (HorseID), DOB, Horse Colour, Horse name. When the Horse comes to the MTU Thoroughbreds clinic, it undergoes an examination by a member of the Veterinary staff. The examination may result in the Horse being prescribed with one or more treatments. Each consultation has a unique consultation number (consultNo) and each type of treatment has a unique treatment number (treatNo).

You should answer all questions that follow. Many attributes are listed, but you may discover that other attributes are also required to complete your design. You should include these attributes.

Task 1 - Data Modelling (E-R Diagrams)

- (a) Identify and list the entities, relationships and multiplicities for the MT Thoroughbreds clinic chain.

Entities –	Staff
	Clinic
	Owner
	Horse
	Consultation
	Treatment

Relationships –

Staff – Clinic (**Manages**)
Staff – Clinic (**Works at**)
Horse – Clinic (**Registers**)
Owner – Horse (**Owne**s)
Staff – Consultation (**Performs**)
Horse – Consultation (**Undergoes**)
Staff – Horse (**Treats**)
Horse – Treatment (**Receives Treatment**)
Consultation – Treatment (**Includes**)

Multiplicities –

Staff – Clinic (**Manages**) – One to One
A staff member can manage only one clinic
A clinic is managed by only one staff member
Staff – Clinic (**Works at**) – One to Many
A clinic can have many staff members
A staff member works at one clinic
Horse – Clinic (**Registers**) – One to Many
A horse is registered at one clinic only
A clinic can have many horses registered
Owner – Horse (**Owne**s) – One to Many
An owner can own many horses
A horse has only one owner
Staff – Consultation (**Performs**) – One to Many
A staff member can perform many consultations
A consultation is performed by only one staff member
Horse – Consultation (**Undergoes**) – One to Many
A horse can have many consultations
A consultation has only one horse
Staff – Horse (**Treats**) – Many to Many
A staff member can treat many horses
A horse can be treated by many staff members
Horse – Treatment (**Receives Treatment**) – Many to Many
A horse can receive many treatments
A treatment can be given to many horses
Consultation – Treatment (**Includes**) – Many to Many
A consultation can include many treatments
A treatment can be included in many consultations

- (b) List and associate all attributes with the appropriate entities and relationships.
Including all primary and foreign keys.

Entities – Staff

Primary Key – staffNo

Foreign Key – clinicNo (Clinic Table)

Attributes – first name, last name, address, DOB, salary, position

Entities – Clinic

Primary Key – clinicNo

foreign Key – staffNo (Staff Table)

Attributes – name, address, phone number

Entities – Owner

Primary Key – ownerNo

Attributes – first name, last name, address, phone number,

Entities – Horse

Primary Key – horseID

Foreign Key – ownerNo (Owner Table), clinicNo (Clinic Table)

Attributes – DOB, colour, name

Entities – Consultation

Primary Key – consulNo

Foreign key – horseID (Horse Table), staffNo (Staff Table)

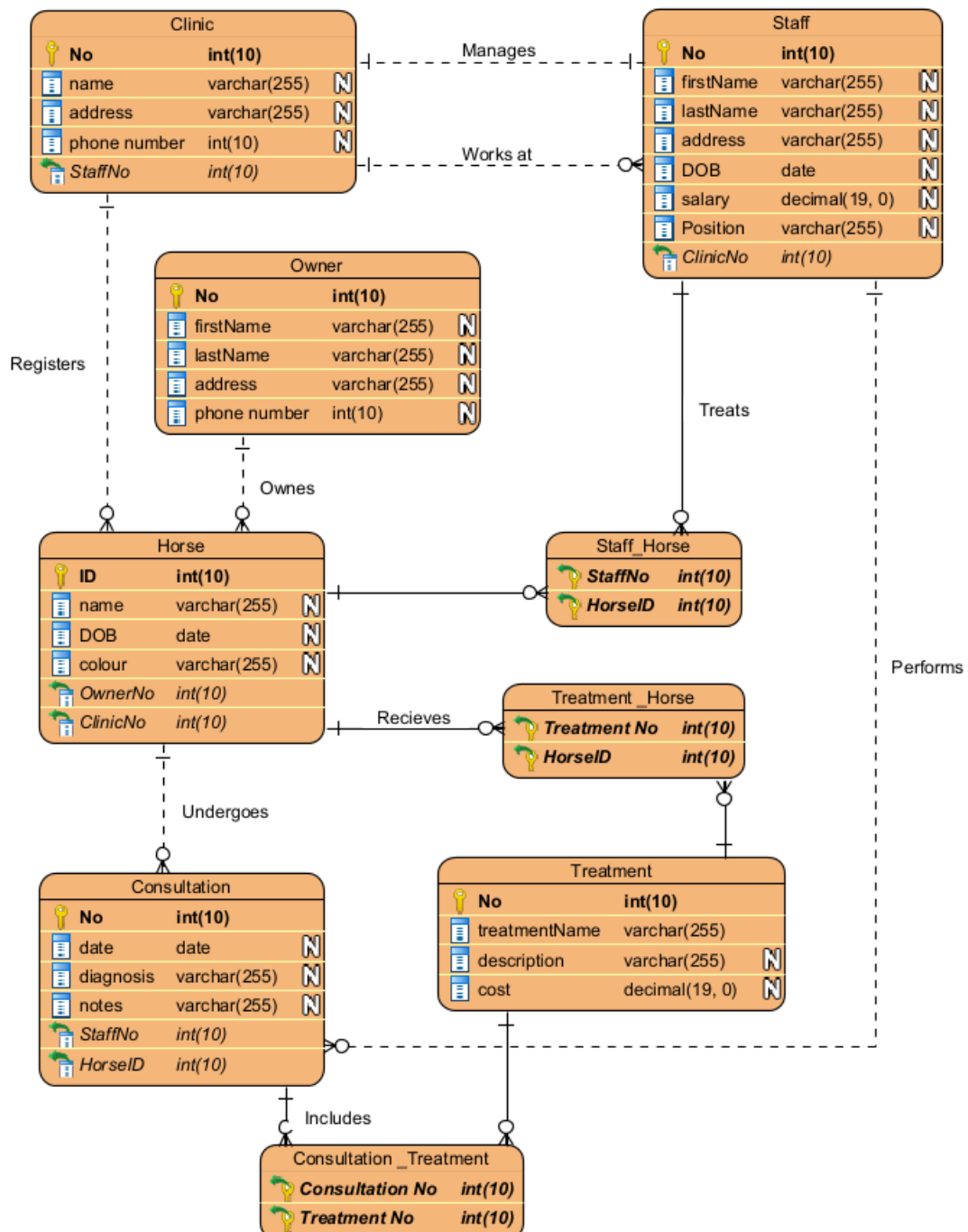
Attributes – date, diagnosis, notes

Entities – Treatment

Primary Key – treatNo

Attributes – name, description, cost

- (c) Using your answers to parts (a) and (b) attempts to represent the data requirements of the MTU Thoroughbreds Clinics as a single ER diagram. You could use a diagram tool like Visual Paradigm to complete this task. Incorporate an image of the diagram into your assignment submission.



To represent the data requirements of MTU Thoroughbreds Clinic in a single ER diagram, I first identified the main tables that stores information for the clinic.

- Staff Table – this table holds information about the clinic’s staff, including their name, contact details, positions and which clinic they are working at.
- Clinic Table – this table holds information about the clinics, including the clinic number, name, location, phone number and the staff member managing the clinic.
- Owner Table – this table holds information about the horses, including their unique identifier, name, date of birth, colour and which clinic they are registered with, as well as their owner.
- Consultation Table – this table records each consultation that a horse undergoes, including the consultation number, date, diagnosis and the staff member performing the consultation.
- Treatment Table – this table contains details about treatments prescribed during consultations, including the treatment number, name, description and cost.

I created relationships between these tables to represent how the different entities interact with each other

- Consultation Table connects to Staff and Horse tables.
A consultation is performed by a staff member. (one to many relationship)
A consultation involves one horse (one to many relationship)
- Horse Table connects to Clinic and Owner tables.
A horse is registered at one clinic. (one to many relationship)
A horse is owned by own owner. (one to many relationship)
- Treatment Table connects to Consultation and Horse tables
A consultation can include many treatments (many to many relationship)
A horse can receive many treatments (many to many relationship)

Each table has a primary key, which uniquely identify each record within the table.

- Staff Table Primary Key – staffNo (Staff Number)
- Clinic Table Primary Key – clinicNo (Clinic Number)
- Owner Table Primary Key – ownerNo (Owner Number)
- Horse Table Primary Key – horseID (Horse Identification)
- Consultation Table Primary Key – consulNo (Consultation Number)
- Treatment Table Primary Key – treatNo (Treatment Number)

Foreign keys are used to link the tables and maintain the relationship between the entities.

- Staff Table Foreign Key – clinicNo (Clinic Table)
- Clinic Table Foreign Key – staffNo (Staff Table)
- Horse Table Foreign Key – ownerNo (Owner Table)
– clinicNo (Clinic Table)
- Consultation Table Foreign Key – staffNo (Staff Table)
– horseID (Horse Table)

Assumptions made while creating the ER diagram

- Each entity (staff, clinic, owner, horse, consultation and treatment) has a unique identifier.
- The foreign keys in the Horse table assume that a horse must be both owned by an owner and registered at a clinic
- The foreign keys in the Consultation table assume that each consultation must involve one horse and be performed by one staff member.
- It is assumed that the tables are filled with complete data to avoid any issues. For Example, every horse must have an owner, and every consultation must have a staff member performing it.

Task 2 – Normalisation and Database Design

- (d) Verify that the ER model drawn in part (c) adheres to the process of normalization through its various stages to BCNF. Clearly identify the correct stage of normalization for each table. You should also carry out normalization that may be necessary to ensure all tables are in BCNF.

Normalization Process

1. **First Normal Form** – The tables are in 1NF because all attributes contain atomic values.
2. **Second Normal Form** – The tables are in 2NF because there is no composite key, so no partial dependencies exist.
3. **Third Normal Form** – The tables are in 3NF because there are no transitive dependencies
4. **Boyce-Codd Normal Form** – The table are in BCNF because every functional dependency has a superkey as the determinant.

All tables are in Boyce-Codd Normal Form because,

- There are no partial dependencies, all functional dependencies depend entirely on the primary key.
- All non-prime attributes are fully functionally dependent on the whole key.
- Every functional dependency's left-hand side is a superkey in each table.

Staff Table							
staffNo	First Name	Last Name	Address	Date of Birth	Salary	Position	clinicNo

Clinic Table				
clinicNo	Name	Address	Phone Number	staffNo

Owner Table				
ownerNo	First Name	Last Name	Address	Phone Number

Horse Table					
horseID	Name	Date of Birth	Colour	ownerNo	clinicNo

Consultation Table					
<i>consulNo</i>	Date	Diagnosis	Notes	<i>horseID</i>	<i>staffNo</i>

Treatment Table			
<i>treatNo</i>	Name	description	cost

Treatment_Horse Table	
<i>treatNo</i>	<i>horseID</i>

Staff_Horse Table	
<i>staffNo</i>	<i>horseID</i>

Consultation_Treatment Table	
<i>treatNo</i>	<i>consulNo</i>

- (e) Once you are satisfied with your design in part (c) & (d), write SQL CREATE TABLE statements to create each table in the MySQL DBMS with all constraints included. You should complete this as one SQL script. Once the tables are created, write one SQL script that inserts at least two records into each table.

SQL CREATE TABLE statements.

```
1 • USE assignment2;
2
3 • CREATE TABLE Clinic
4   (clinicNo INT NOT NULL,
5    name VARCHAR(255),
6    address VARCHAR(255),
7    phoneNumber VARCHAR(255),
8    PRIMARY KEY (clinicNo));
9
10 • CREATE TABLE Staff
11   (staffNo INT NOT NULL,
12    firstName VARCHAR(255),
13    lastName VARCHAR(255),
14    address VARCHAR(255),
15    DOB DATE,
16    salary DECIMAL(10,2),
17    position VARCHAR(255),
18    clinicNo INT,
19    PRIMARY KEY (staffNo),
20    FOREIGN KEY (clinicNo) REFERENCES Clinic(clinicNo));
21
22 • CREATE TABLE Owner
23   (ownerNo INT NOT NULL,
24    firstName VARCHAR(255),
25    lastName VARCHAR(255),
26    address VARCHAR(255),
27    phoneNumber VARCHAR(255),
28    PRIMARY KEY (ownerNo));
29
30 • CREATE TABLE Horse
31   (horseID INT NOT NULL,
32    name VARCHAR(255),
33    DOB DATE,
34    colour VARCHAR(255),
35    ownerNo INT,
36    clinicNo INT,
37    PRIMARY KEY (horseID),
38    FOREIGN KEY (ownerNo) REFERENCES Owner(ownerNo),
39    FOREIGN KEY (clinicNo) REFERENCES Clinic(clinicNo));
40
41 • CREATE TABLE Treatment
42   (treatNo INT NOT NULL,
43    treatmentName VARCHAR(255),
44    description TEXT,
45    cost DECIMAL(10, 2),
46    PRIMARY KEY (treatNo));
47
48 • CREATE TABLE Consultation
49   (consulNo INT NOT NULL,
50    date DATE,
51    diagnosis TEXT,
52    notes TEXT,
53    horseID INT,
54    staffNo INT,
55    PRIMARY KEY (consulNo),
56    FOREIGN KEY (horseID) REFERENCES Horse(horseID),
57    FOREIGN KEY (staffNo) REFERENCES Staff(staffNo));
58
59 • CREATE TABLE Consultation_Treatment
60   (consulNo INT NOT NULL,
61    treatNo INT NOT NULL,
62    PRIMARY KEY (consulNo, treatNo),
63    FOREIGN KEY (consulNo) REFERENCES Consultation(consulNo),
64    FOREIGN KEY (treatNo) REFERENCES Treatment(treatNo));
65
66 • CREATE TABLE Treatment_Horse
67   (treatNo INT NOT NULL,
68    horseID INT NOT NULL,
69    PRIMARY KEY (treatNo, horseID),
70    FOREIGN KEY (treatNo) REFERENCES Treatment(treatNo),
71    FOREIGN KEY (horseID) REFERENCES Horse(horseID));
72
73 • COMMIT;
```

SQL INSERT script

```
1      USE assignment2;
2
3      ● INSERT INTO Clinic VALUES(1, 'The Kingdom Clinic', 'Market St, Kenmare, County Kerry', '123456789');
4      ● INSERT INTO Clinic VALUES(2, 'The Rebel Clinic', 'Tramore Road, County Cork', '987654321');
5
6      ● INSERT INTO Staff VALUES(1, 'Mary', 'Lynch', '12 Green St, Kildare, County Kerry', '1998-05-14', 50000, 'Vet, Manager', 1);
7      ● INSERT INTO Staff VALUES(2, 'John', 'Brown', '4 Blossom Tree Lane, Tralee, County Kerry', '1990-03-22', 40000, 'Vet', 1);
8      ● INSERT INTO Staff VALUES(3, 'Lilly', 'White', '12 Douglas Road, County Cork', '1983-07-30', 45000, 'Vet, Manager', 2);
9
10     ● INSERT INTO Owner VALUES(1, 'Mark', 'Collins', '1 Light Street, County Kerry', '0876543210');
11     ● INSERT INTO Owner VALUES(2, 'Deidre', 'Long', '2 Patricks Street, Cork', '0865432109');
12
13     ● INSERT INTO Horse VALUES(1, 'Sherlock', '2019-02-12', 'Black', 1, 1);
14     ● INSERT INTO Horse VALUES(2, 'Snow', '2017-06-25', 'White', 2, 2);
15
16     ● INSERT INTO Treatment VALUES(1, 'Vaccination', 'A vaccine', 100.00);
17     ● INSERT INTO Treatment VALUES(2, 'Hoof Care', 'hoof care and trimming', 75.00);
18     ● INSERT INTO Treatment VALUES(3, 'Dental Check', 'Examination and care of the horse's teeth', 120.00);
19
20     ● INSERT INTO Consultation VALUES(1, '2024-11-01', 'Mild fever', 'Requires observation and treatment', 1, 1);
21     ● INSERT INTO Consultation VALUES(2, '2024-11-02', 'Leg injury', 'Requires bandaging and rest', 2, 2);
22
23     ● INSERT INTO Consultation_Treatment VALUES(1, 1);
24     ● INSERT INTO Consultation_Treatment VALUES(1, 2);
25     ● INSERT INTO Consultation_Treatment VALUES(2, 3);
26
27     ● INSERT INTO Treatment_Horse VALUES(1, 1);
28     ● INSERT INTO Treatment_Horse VALUES(2, 1);
29     ● INSERT INTO Treatment_Horse VALUES(3, 2);
30
31     ● COMMIT;
```

Staff Table

staffNo	firstName	lastName	address	DOB	salary	position	clinicNo
1	Mary	Lynch	12 Green St, Kildare, County Kerry	1998-05-14	50000.00	Vet, Manager	1
2	John	Brown	4 Blossom Tree Lane, Tralee, County Kerry	1990-03-22	40000.00	Vet	1
3	Lilly	White	12 Douglas Road, County Cork	1983-07-30	45000.00	Vet, Manager	2

Clinic Table

clinicNo	name	address	phoneNumber
1	The Kingdom Clinic	Market St, Kenmare, County Kerry	123456789
2	The Rebel Clinic	Tramore Road, County Cork	987654321

Owner Table

ownerNo	firstName	lastName	address	phoneNumber
1	Mark	Collins	1 Light Street, County Kerry	0876543210
2	Deidre	Long	2 Patricks Street, Cork	0865432109

Horse Table

horseID	name	DOB	colour	ownerNo	clinicNo
1	Sherlock	2019-02-12	Black	1	1
2	Snow	2017-06-25	White	2	2

Consultation Table

consulNo	date	dignosis	notes	horseID	staffNo
1	2024-11-01	Mild fever	Requires observation and treatment	1	1
2	2024-11-02	Leg injury	Requires bandaging and rest	2	2

Treatment Table

treatNo	treatmentName	description	cost
1	Vaccination	A vaccine	100.00
2	Hoof Care	hoof care and trimming	75.00
3	Dental Check	Examination and care of the horse's teeth	120.00

Task 3 – Physical Database Design

- (f) Explain the difference between a primary key and a surrogate key. In your database design, where would it be appropriate to use a surrogate key, and why?

A Primary key is a column that uniquely identifies each row in a table. The values of a primary key must be unique and not null. For example, in the Staff Table, staffNo is the primary key because it uniquely identifies each staff member.

A surrogate key is an artificial, system-generated key used to uniquely identify a row in a table when a natural key is either too complex or could change. It is usually an auto-incrementing number. For example, in the Horse Table, instead of using a horse's name or registration number as a key, we use a surrogate key like horseID.

In summary the difference between a primary and surrogate key is that a primary key uses real-world data to uniquely identify rows, while a surrogate key is an artificial key used when natural keys are too complicated or likely to change.

In my database design, a surrogate key is appropriate in the Horse Table because, a horse's name or registration number might change, making them unreliable as keys. A surrogate key like horseID gives a simple, stable identifier for the horse.

- (g) Identify one or more columns in your table where it would be beneficial to create an index. Explain why you have chosen these columns for indexing and describe the impact it would have on query performance. Consider both read and write operations.

In the Staff Table it would be beneficial to create an index on staffNo, because it is the primary key and is frequently used to join with other tables. Indexing staffNo would speed up queries that search for or link staff data. The impact on query performance would be an improvement in read performance, particularly when searching for staff or joining tables on staffNo. The downside is that there might be a slight performance degradation for write operations like inserting or updating data, as the index must be updated whenever staffNo is changed.

In the Consultation table it would be beneficial to create an index on horseID, as this column is often used to filter consultations by a specific horse. Indexing horseID would improve query performance by speeding up searches for consultations based on a particular horse. Like the staff table, this would enhance read performance, but might slightly impact write operations, as the index would need to be updated when data changes.

In the Horse table it would be beneficial to create an index on clinicNo, as this column is used to find all horses at a specific clinic. Indexing clinicNo would improve read performance for queries filtering by clinic. As with the other indexes, the downside would be a small performance cost for write operations due to the need to maintain the index.

By indexing these three columns staffNo, horseID and clinicNo, retrieving data would be significantly faster, especially for queries filtering by these attributes. However, the trade-off is that write operations might experience a slight slowdown due to the overhead of maintaining the indexes.

- (h) Explain the concept of clustering in database design. How does clustering help improve query performance in relational databases? In the above design of the MTU Thoroughbreds Clinic database, which tables (or pairs of tables) would benefit from clustering? Justify your answer with respect to data access patterns and potential performance improvements.

Clustering in database design means storing related data close together on the disk. This helps speed up queries because the database doesn't have to search through the entire disk for the data. Instead, it can quickly find everything it needs since the related data is stored next to each other.

For example, if you often look for consultations for a particular horse, clustering the Consultation Table by horseID would make sense. All consultations for the same horse would be stored together, so when you search for them, the database can quickly retrieve the data without having to look all over the disk.

The benefits of clustering include

Faster Queries, when related data is stored together, the database can find it quicker, this reduces the time spend reading from the disk.

Better Disk Usage, since the data is stored together, it's easier and faster to retrieve. The disk space is used more effectively, reducing the time spent retrieving the data.

Table that would benefit from clustering,

Consultation Table and Horse Table –

Clustering the Consultation table by horseID would speed up queries that find all consultations for a specific horse. Since all consultations for the same horse are stored together, the database can quickly access the data and reduce the number of disk reads.

Horse Table and Owner Tables –

Clustering the Horse table by ownerNo would store all horses owned by the same person together. This would speed up searches for all horses owned by a particular person, as the related data would be stored close to each other.